TRACCIA 2

Si studi la funzione

$$f(x) = \frac{x^2}{\ln x - 1}$$

DOMINIO

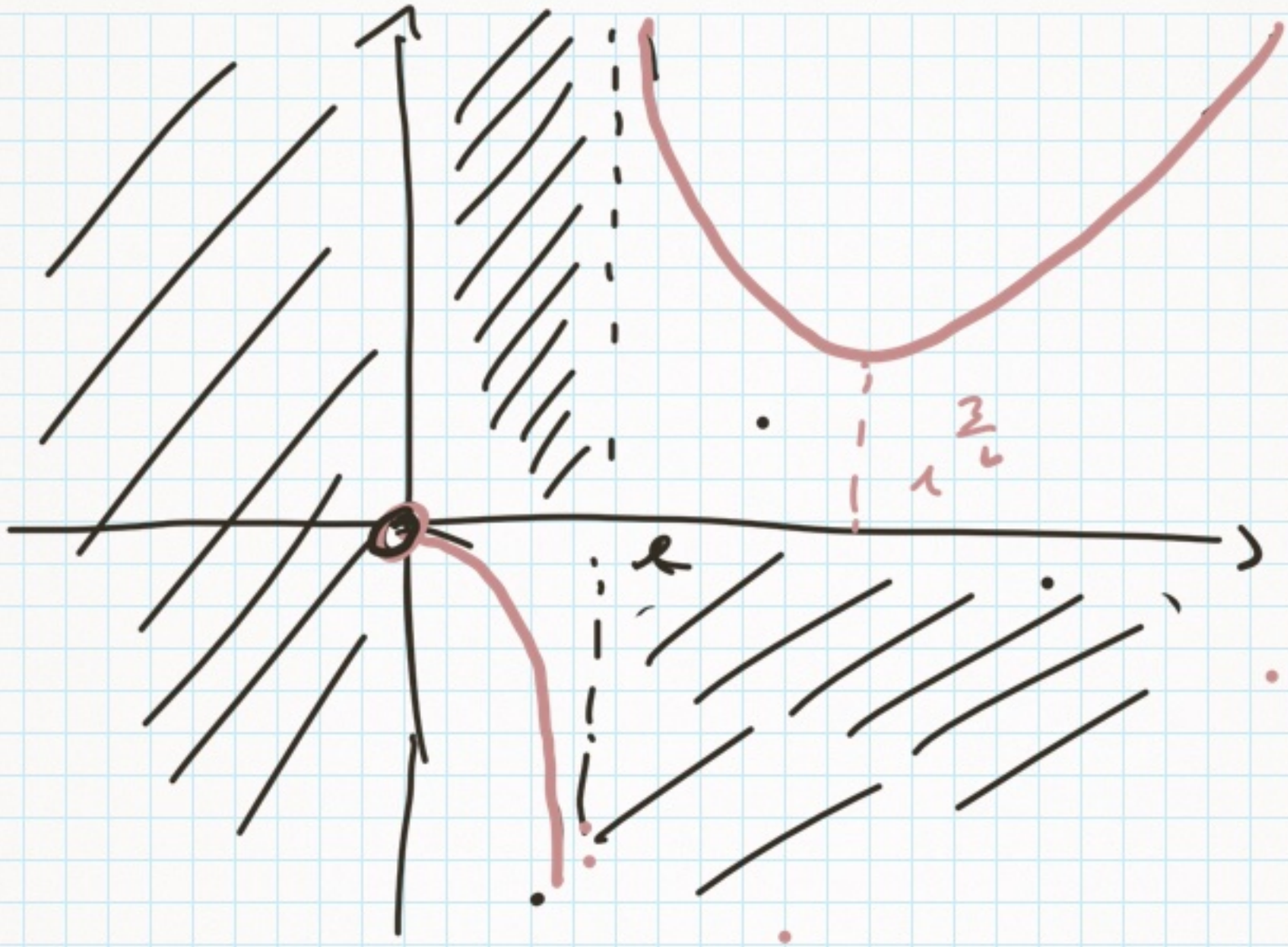
$$\begin{cases} x > 0 & \text{(il logaritmo deve essere definito)} \\ \ln x - 1 \neq 0 & \text{(denominatore} \neq 0\text{)} \end{cases}$$

$$\ln x \neq 1$$

$$x \neq e^1 = e$$

Dominio:

$$]0, +\infty[\setminus \{e\}$$



- DOMINIO NON SIMMETRICO
  f non è né pari né dispari
- $x=0 \notin D \Rightarrow$ non abbiamo intersezioni con y

INTERSEZIONE ASSE X E POSITIVITÀ

$$f(x) \geq 0$$

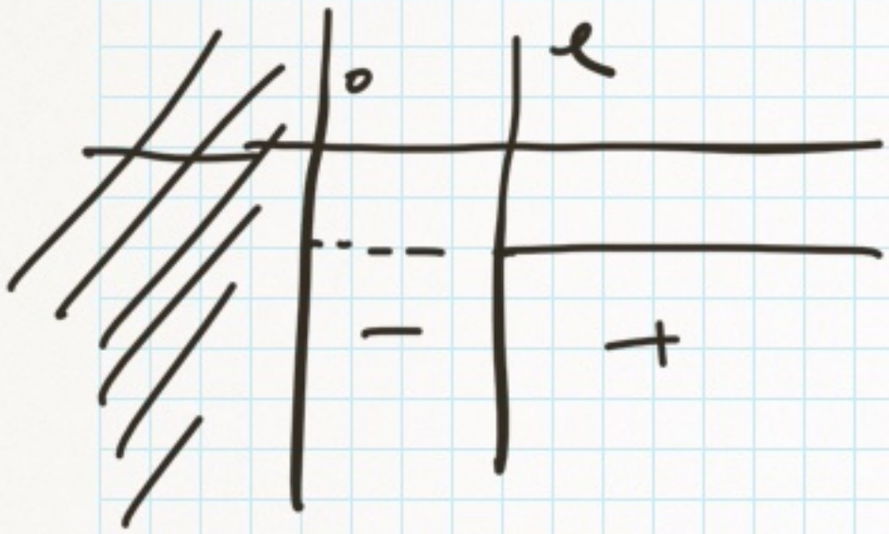
$$\frac{x^2}{\ln x - 1} \geq 0$$

$x^2 \geq 0 \quad \forall x \in \mathbb{R}$, in particolare

$x^2 = 0 \iff x = 0 \notin D$

$f(x) \neq 0$

$\ln x - 1 > 0 \iff \ln x > 1 \iff x > e$



$f(x) > 0 \iff x > e$

$f(x) < 0 \iff 0 < x < e$

ASINTOTI

$$\lim_{x \to 0^+} \frac{x^2}{\ln x - 1} = \frac{0}{\ln 0^+ - 1} = \frac{0}{-\infty - 1} =$$

$$= \frac{0}{-\infty} = 0$$

$$\lim_{x\to e^-} \frac{x^2}{\ln x - 1} = \frac{e^2}{1-1} = \frac{e^2}{0^-} = -\infty$$

$x = e$ ASINTOTO VERTICALE SINISTRO

$$\lim_{x\to e^+} \frac{x^2}{\ln x - 1} = \frac{e^2}{0^+} = +\infty$$

$x = e$ ASINTOTO VERS. DESTRO

ASINTOTO ORIZZONTALE

$$\lim_{x \to +\infty} \frac{x^2}{\ln x - 1} = +\infty$$

perché x^2 va a $+\infty$ più velocemente di $\ln x$

ASINTOTO OBLIQUO

$$\lim_{x \to +\infty} \frac{f(x)}{x} = \lim_{x \to +\infty} \frac{x^2}{(\ln x - 1)x} =$$

$$= \lim_{x \to +\infty} \frac{x}{\ln x - 1} = +\infty$$

NO ASINTOTO OBLIQUO

DERIVABILITÀ E MONOTONIA

$$f(x) = \frac{x^2}{\ln x - 1}$$

$$f'(x) = \frac{2x(\ln x - 1) - x^2 \cdot \frac{1}{x}}{(\ln x - 1)^2} =$$

$$= \frac{2x(\ln x - 1) - x}{(\ln x - 1)^2} = \frac{x(2\ln x - 2 - 1)}{(\ln x - 1)^2}$$

$$= \frac{x(2\ln x - 3)}{(\ln x - 1)^2}$$

LA FUNZIONE È DERIVABILE NEL SUO DOMINIO

$f'(x) \geq 0$ (STUDIO MONOTONIA)

> 0 welodamo.

$$f'(x) = \frac{x(2\ln x - 3)}{(\ln x - 1)^2} \geq 0 \iff$$

> 0

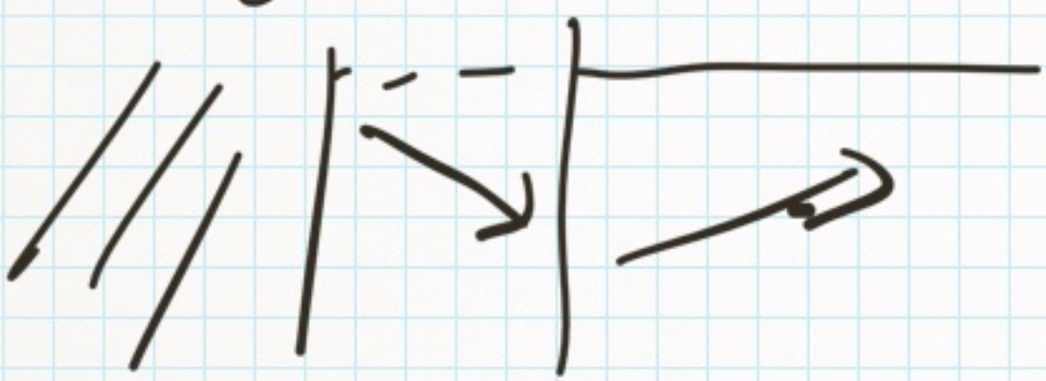
$$2\ln x - 3 \geq 0 \iff \ln x \geq \frac{3}{2}$$

$$\ln x \quad \frac{3}{2}$$

$$\iff e \geq e \iff x \geq e^{\frac{3}{2}}$$

$$e^{\frac{3}{2}}$$

0



$x = e^{\frac{3}{2}}$ punto di minimo

CONVESSITÀ

$$f'(x) = \frac{x(2\ln x - 3)}{(\ln x - 1)^2}$$

$$f''(x) = \frac{\left(2\ln x - 3 + x \cdot \frac{2}{x}\right)(\ln x - 1)^2}{(\ln x - 1)^4}$$

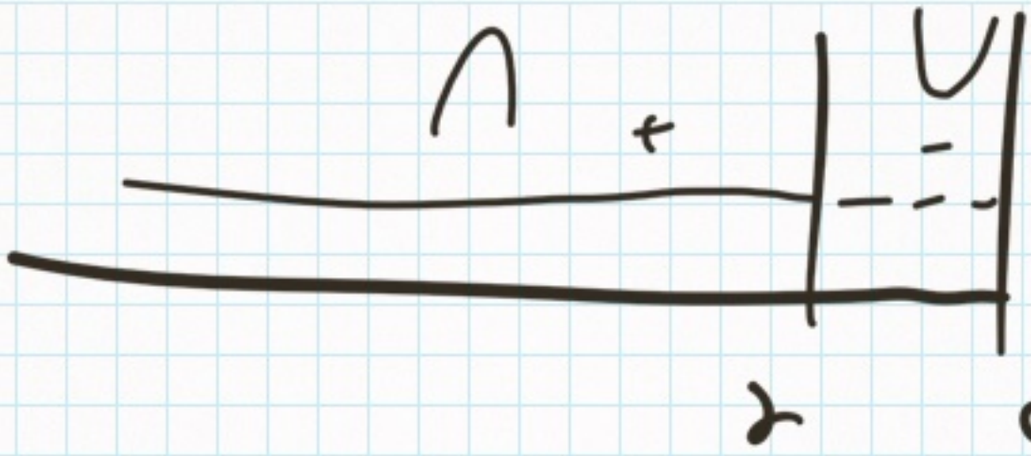
$$- \frac{x(2\ln x - 3)\, 2(\ln x - 1)\cdot \frac{1}{x}}{(\ln x - 1)^4} = \quad \cdots$$

$$= \frac{(2\ln x - 1)(\ln x - 1)^2 - 2(2\ln x - 3)(\ln x - 1)}{(\ln x - 1)^4} =$$

$$= \frac{(\ln x - 1)\left[(2\ln x - 1)(\ln x - 1) - 2(2\ln x - 3)\right]}{(\ln x - 1)^{4}\ 3}$$

$$= \frac{2\ln^2 x - 2\ln x - \ln x + 1 - 4\ln x + 6}{(\ln x - 1)^3} =$$

$$= \frac{2\ln^2 x - 7\ln x + 7}{(\ln x - 1)^3} \geq 0$$



$x > r \iff$

$\cdot \; (\lambda x - 1) > 0 \iff \lambda x - 1 > 0$ (3)

$\Delta = (-2)^2 - 4 \cdot 1 \cdot 2 = 4 - 8 < 0$

$\forall t \in \mathbb{R} \quad t^2 - 2t + 2 \geq 0$

$t = \lambda x$

$\cdot \; \lambda^2 x^2 - 2\lambda x + 2 \geq 0$

$\forall x \in A \quad x > 0$

Si dica per quali $x \in \mathbb{R}$ la serie

$$\sum_{n=0}^{+\infty} \left(\frac{x^2-2}{x+4} \right)^n$$

converge e si calcoli la somma.

Sia $q = \frac{x^2-2}{x+4}$

$\sum_{n=0}^{+\infty} q^n$ converge se e solo se

$|q| < 1 \iff -1 < q < 1$

e la somma è $\frac{1}{1-q}$

Occorre risolvere

$$-1 < \frac{x^2-2}{x+4} = y < 1$$

$$\begin{cases} \frac{x^2-2}{x+4} < 1 & (1) \\ \frac{x^2-2}{x+4} > -1 & (2) \end{cases}$$

(1) $\frac{x^2-2}{x+4} - 1 < 0 \Rightarrow \frac{x^2-x-6}{x+4} < 0$

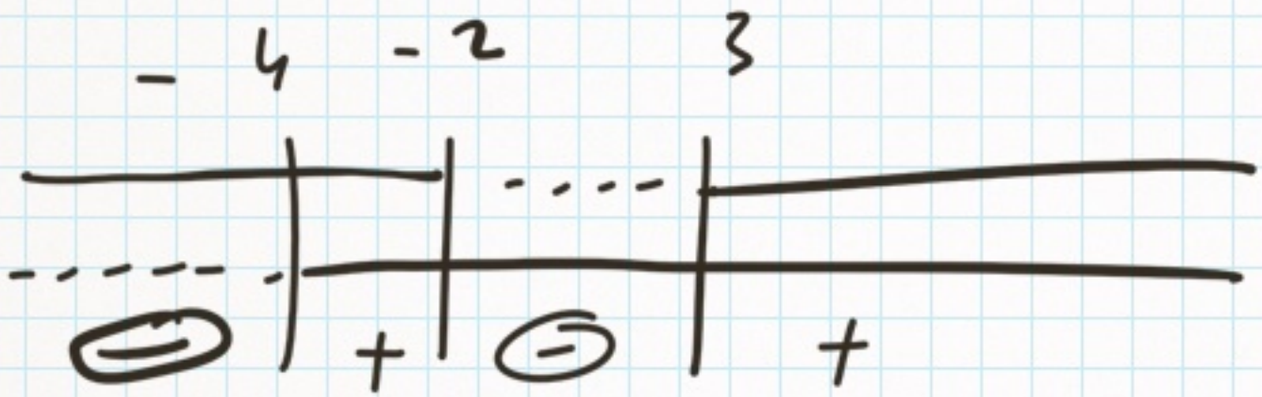
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- $x^2 - x - 6 > 0 \iff x < -2 \vee x > 3$

$x^2 - x - 6 = 0$

$x_{1,2} = \frac{1 \pm \sqrt{1+24}}{2} = \frac{1 \pm 5}{2} \begin{matrix} 3 \\ -2 \end{matrix}$

- $x + 4 > 0 \iff x > -4$.



$x < -4 \quad \vee \quad -2 < x < 3$

$$x+y>0 \iff x>-y$$

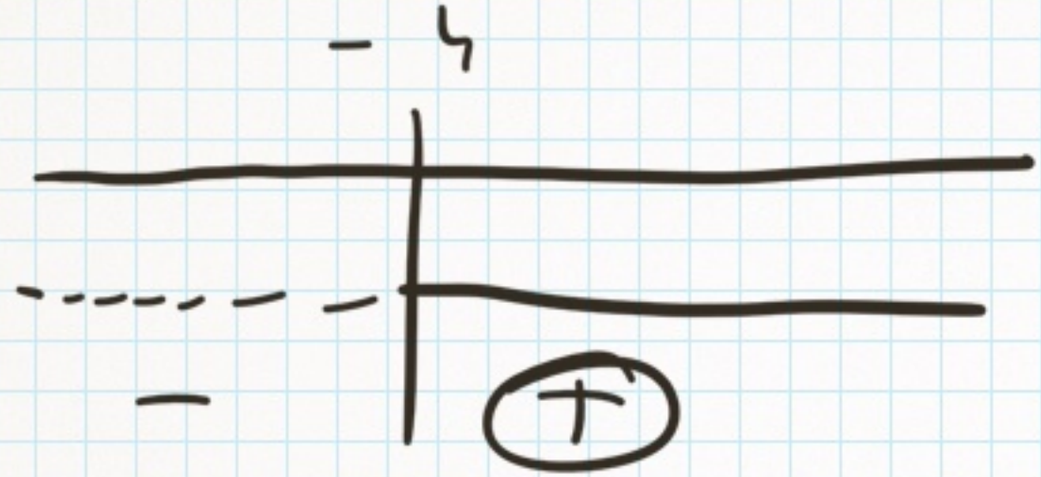
$$\Delta = 1-4 = -3 < 0$$

$$\forall x \in \mathbb{R} \quad x^2+x+1>0$$

$$\frac{x+y}{x^2+x+2} > 0$$

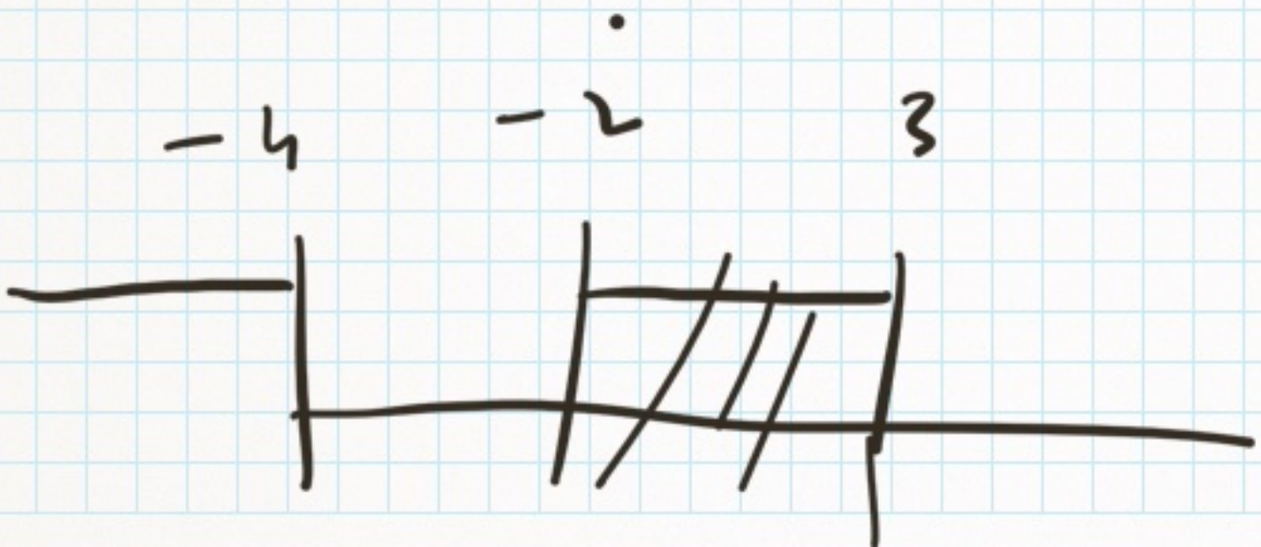
$$\frac{x+y}{-x^2+x+y-2} < 0$$

$$1+\frac{x+y}{2-x^2} > 0 \qquad (1)$$



$x > -4$

$$\begin{cases} x < -4 \text{ v } -2 < x < 3 & ① \\ x > -4 & ② \end{cases}$$



La serie converge se

$$-2 < x < 3$$ e ha per somma

$$s = \frac{1}{1 - \frac{x^2-2}{x+3}}$$

ESEMPIO COMUNE

Si usa il criterio della radice in

altri studiare se l'ordine converge

$$\lim_{n\to+\infty} \sqrt[n]{\left|\frac{x^2-2}{x+3}\right|^n} = \lim_{n\to+\infty} \left|\frac{x^2-2}{x+3}\right| =$$

$$\left|\frac{x^2-2}{x+4}\right| < 1$$

Si calcola il seguente integrale

$$\int \frac{\ln x+1}{x(\ln^2 x+3)}\,dx \qquad \ln x = t$$

$$= \int \frac{t+1}{t^2+3}\,dt = \frac{1}{2}\int \frac{2t}{t^2+3}\,dt + \int \frac{1}{t^2+3}\,dt$$

$$= \frac{1}{2} \ln(t^2+3) + \frac{1}{3}\int \frac{1}{\left(\frac{t}{\sqrt{3}}\right)^2+1}\, dt$$

$$= \frac{1}{2} \ln(t^2+3) + \frac{1}{\sqrt{3}}\int \frac{\frac{1}{\sqrt{3}}}{\left(\frac{t}{\sqrt{3}}\right)^2+1}\, dt$$

$$= \frac{1}{2} \ln(t^2+3) + \frac{1}{\sqrt{3}} \operatorname{arctg}\left(\frac{t}{\sqrt{3}}\right) + c$$

$t = \ln x$

$$= \frac{1}{2} \ln(\ln^2 x + 3) + \frac{1}{\sqrt{3}} \operatorname{arctg}\left(\frac{\ln x}{\sqrt{3}}\right) + c$$

. $\lim_{x\to-\infty} \sqrt{x^2-3x+2} + x = +\infty - \infty$

$= x - x = 0$

ВЗЯВ NAC, ТЗАДЕ

$$= \lim_{x\to-\infty} \frac{\left(\sqrt{x^2-3x+2} + x\right)\left(\sqrt{x^2-3x+2} - x\right)}{\sqrt{x^2-3x+2} - x}$$

$$= \lim_{x\to-\infty} \frac{x^2 - 3x + 2 - x^2}{\sqrt{x^2} - x} =$$

$$= \lim_{x\to-\infty} \frac{-3x+2}{|x| - x} =$$

$\sqrt{x^2} = |x|$

. .

$$= \lim_{x \to -\infty} \frac{-3x+2}{-x-x} =$$

$$= \lim_{x \to -\infty} \frac{-3x}{-2x} = -\frac{3}{2}$$

- $\lim_{x \to 0} \frac{(1-\cos(3x))\arcsin(2x)}{(e^{2x^2}-1)\tan(4x^2)}$

$1-\cos x \sim \frac{1}{2}x^2$ $\quad x \to 0$

$1-\cos(3x) \sim \frac{1}{2}(3x)^2 = \frac{9}{2}x^2$.

$$\text{for } x \sim x \qquad x \to 0$$

$$\text{for } (\dot{x}^2) \sim \dot{x}^2$$

$$\dot{x}$$

$$\frac{2}{2x} \sim 1 - x$$

$$2x$$

$$x \sim 1 - x \,.$$

$$x$$

$$\text{Then } (x^2) \sim 2x$$

$$x \sim x \text{ then } 0 \,.$$

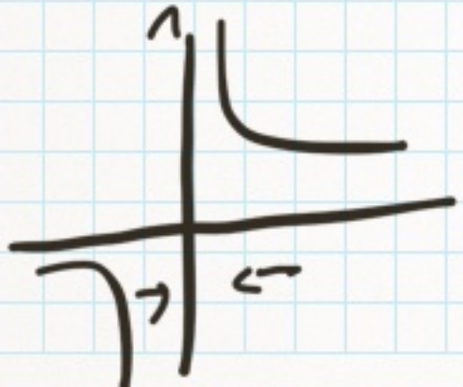
$$0 \leftarrow x \qquad\qquad x \to 0$$

Soluzioni Esercizio 6

$$\lim_{x\to 0} \frac{(1-\cos(3x))\arcsin(2x)}{(e^{2x^2}-1)\tan(4x^3)} =$$

$$= \lim_{x\to 0} \frac{\frac{9}{2}x^2 \cdot 2x}{2x^2 \cdot 4x^3} = \lim_{x\to 0} \frac{9x^3}{8x^5} =$$

$$= \frac{9}{8} \lim_{x\to 0} \frac{1}{x^2}$$



$\lim_{x \to 0^+} \frac{1}{x} = +\infty$ $\lim_{x \to 0^-} \frac{1}{x} = -\infty$

Allora ~~$\exists$~~ $\lim_{x \to 0} \frac{1}{x}$

Si stabilisce n per la funzione

$f(x) = x^3 + 4x + 3 \qquad x \in [-1, 1]$

volgo la ipotesi del teorema degli zeri:

- Dimostrare che lo zero è unico e appartiene a $]-1, -\frac{1}{2}[$

TEOREMA DEGLI ZERI

$f: [a,b] \to \mathbb{R}$ continua t.c.

$f(a) \cdot f(b) < 0$

Allora $\exists x_0 \in]a,b[$ t.c. $f(x_0) = 0$

$f(x) = x^3 + 4x + 3$

polinomio, quindi funzione continua

$[-1,1]$ è chiuso e limitato

$f(-1) = -1 - 4 + 3 = -2 < 0$

$f(1) = 1 + 4 + 3 = 8 > 0$

Allora $f(-1) \cdot f(1) < 0$

le ipotesi sono verificate

$\Rightarrow \exists x_0 \in]-1,1[$ t.c. $f(x_0) = 0$

$x^3 + 4x + 3 = 0$

NON È risolvibile facilmente

la f è STRETTAMENTE MONOTONA
oltre lo zero è unico in questo
f è iniettiva

$f'(x) = 3x^2 + 4 > 0 \Rightarrow$ f è strett. monotona

$f(-1) \cdot f\left(-\frac{1}{2}\right) < 0$

$\Rightarrow x_0 \in \left]-1, -\frac{1}{2}\right[$

(Applico il teorema degli zeri e
f nell'intervallo $\left[-1, -\frac{1}{2}\right]$)